



## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design

#### 1.0 SCOPE

The scope of this document includes design and analysis procedure of grillage and sea fastening components based on reactions or loads calculated in the global module design and configured to ship frame dimensions, relative position and load carrying limits.

The roll and pitch bracing or stoppers need to be incorporated into the global module design, due to their impact on member unity checks, module deflections and redistribution of loads within the structure. The module engineer shall perform checks of the module to meet adequacy requirements of all members and connections considering the impacts of roll / pitch bracing or stoppers on the module design.

The grillage and sea fastening design is typically a deliverable to be reviewed and approved by the MWS, prior to module sail away. It should be noted that the web frame spacing and web frame capacity need to be taken into consideration when the Project is deciding on items such as column spacing. In addition, SPMT placement beneath the module should be dimensioned in consultation with Rigging, to avoid the use of temporary framing that alters the natural frame load path, especially for tall and narrow modules.

#### 2.0 PROCESS AND PROCEDURE

The overall intent is to provide a reliable load path from cargo (module) to ship frame (hard points) that are accessible and constructable at deck. The loads to be considered are attributed to the harbor condition and the sailing condition. In addition, the harbor and sailing conditions have 2 specific sub-conditions each and are detailed as follows:

| Load Case            | Framing Condition                                                                                                                                                                                                                                                                                                                                          | Loading and Strength Consideration                                                                                                                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Case 1<br>(Harbor 1) | Module Frame supported by Grillage only<br>(Immediately after set-down)                                                                                                                                                                                                                                                                                    | Gravity induced loads on transportation mass plus a maximum frictional force on the grillage estimated at 0.20 of the vertical loads. No ship movement and additional load to the ship will occur until 25% of the sea fastening is installed. No ASD overstrength consideration permitted. |
| Case 2<br>(Harbor 2) | Module Frame supported by Grillage and a minimum number of sea-fastening elements to be installed prior to turning of the vessel to be agreed upon with module yard contractor and MWS. Grillage are considered as roller supports (see section 4.2). This is a temporary condition in between module set down and complete installation of sea-fastening. | Gravity induced loads on transportation mass plus a maximum of 10% of transportation mass applied in both principal lateral axes. Minor movement can be done but sailing will not happen until complete installation of sea fastening is done. No ASD overstrength consideration permitted. |



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|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                        |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Case 3<br>(Sailing 1) | Module Frame supported by Grillage + complete sea fastening. Primary lateral resisting system are pinned (pitch and roll braces in most cases) and columns are modelled as a roller. Lateral displacements or forces on column base are zero. The impact of placing the braces (pitch and roll) after setting down where all the weight is supported without the braces should be considered preferably through a 2-stage loading analysis. A suitable software should be chosen as mentioned in 4.5.4.                                                                           | Complete marine transportation loading in accordance with project specifications. One third ASD overstrength consideration permitted in the design of non-G&SF members resisting inertial marine acceleration loading. |
| Case 4<br>(Sailing 2) | Module Frame supported by Grillage + complete sea fastening. Lateral displacements or forces on column base are included. Supports are modelled as pins if sliding does not occur. If sliding occurs, supports are modelled as rollers that are pushed by a frictional force with coefficient of friction $\mu = 0.3$ . The impact of placing the braces (pitch and roll) after setting down where all the weight is supported without the braces should be considered preferably through a 2-stage loading analysis. A suitable software should be chosen as mentioned in 4.5.4. | Complete marine transportation loading in accordance with project specifications. One third ASD overstrength consideration permitted in the design of non-G&SF members resisting inertial marine acceleration loading. |

The module engineer should calibrate with MWS on the analysis considerations specific to the module. It may be acceptable for the MWS to have a flexible module analyzed only for Case 3 loading (considering vertical hog and sag only) and hence, simplify both the module and G&SF design.

In contrast, a rigid frame may induce significant stress due to the horizontal movement from hog and sag and the Case 4 analysis would be necessary. It is also a concern for the ship if it is subjected to a lot of localized reactions. Hence, rigid module framings are to be avoided.

Rigid framing examples are a) multiple braced bays in a singular (vertically co-planar) column grid, b) truss type configurations at any level of the structure and, c) short stubby columns with deep fixed connection beams at the first level.

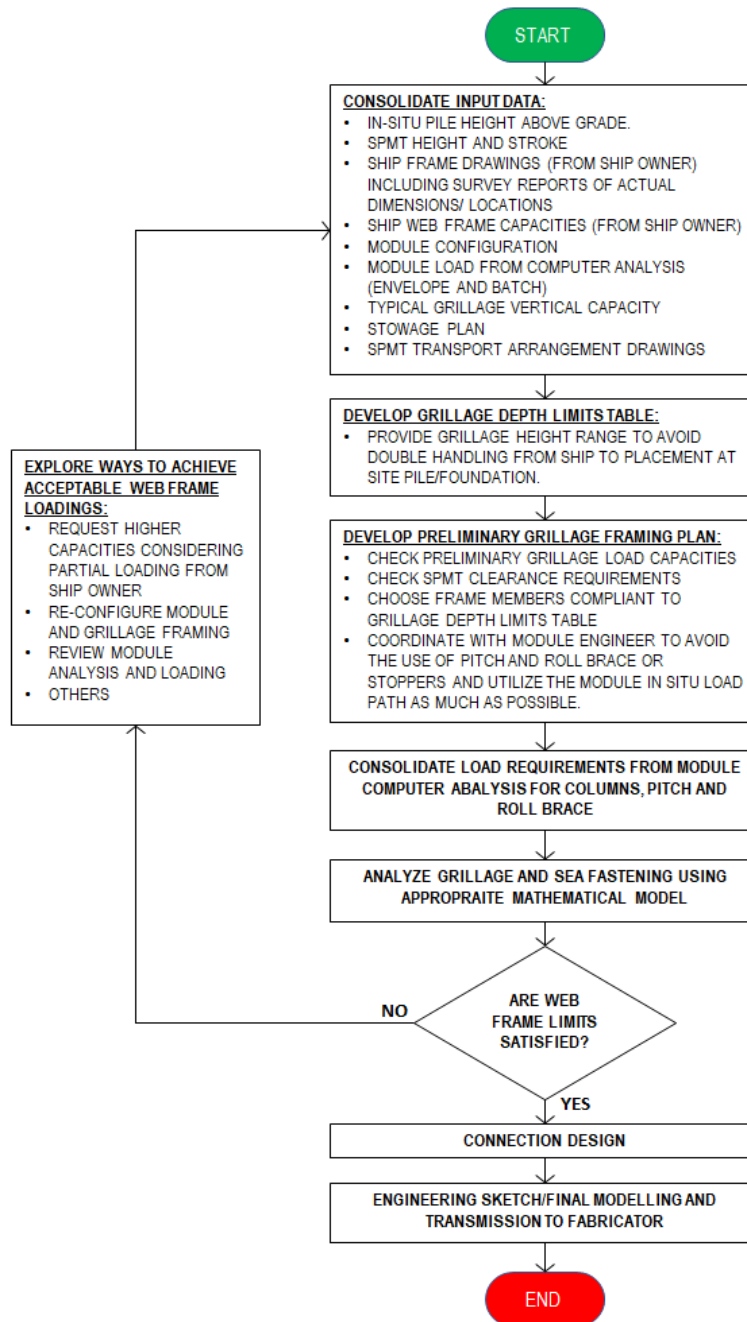
Flexible framing should be used. Examples of flexible framing are as follows: a) 2-way moment frames and b) transverse moment frames coupled with a single bay braced frame (preferably chevron type) in the longitudinal direction.



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#### 2.1 Process Flow





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#### 2.2 Governing Codes

The following codes are typically used together with specific project required documents and references.

- DNVGL-ST-N001 2016
- AISC 360-16
- AISC Steel Construction Manual 2019
- Note that for the transitory nature of marine acceleration loads, DNVGL-ST-N001 2016 allows a 1/3 increase in strength if Limit State 2 (LS2) load combinations and is typically used considering AISC ASD considerations.

#### 2.3 Grillage and Sea fastening configurations (Typical design)

Ship deck where the grillage and sea fastening components are installed is subject to deformation from hog and sag encountered during sea-voyage. As such, grillage configuration should be able to deform at deck connections without significant increase in internal stress between grillage components. A two-point support configuration coupled with pin connections between beams are typically used. This concept is further explained graphically below:

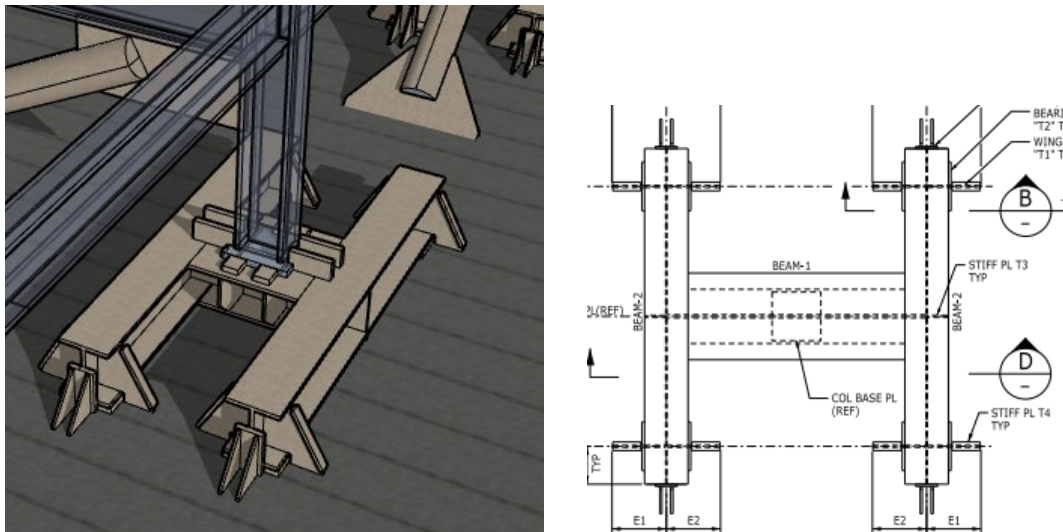


Figure 1 – 4-Point Grillage



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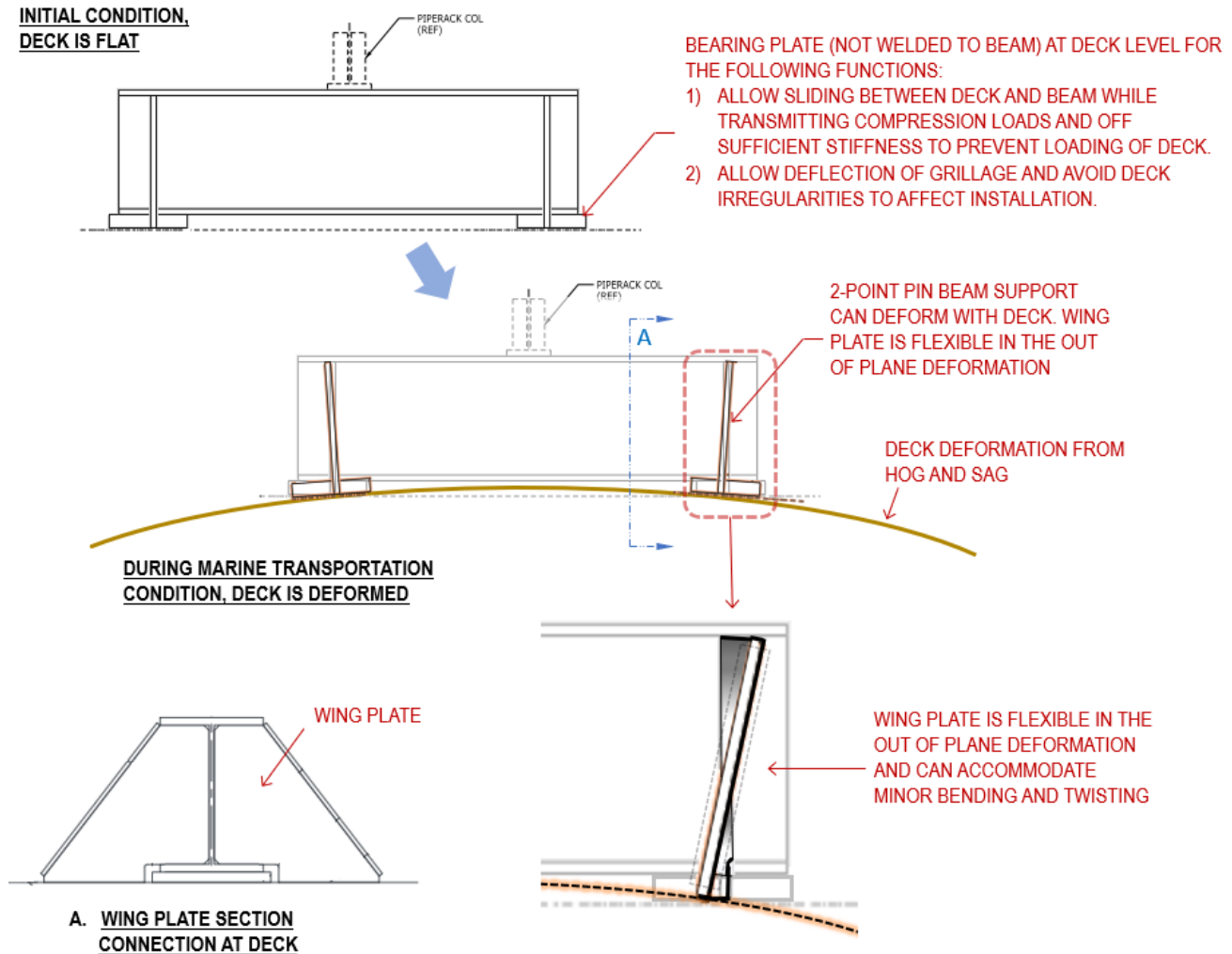


Figure 2 – 4-Point Grillage flexibility



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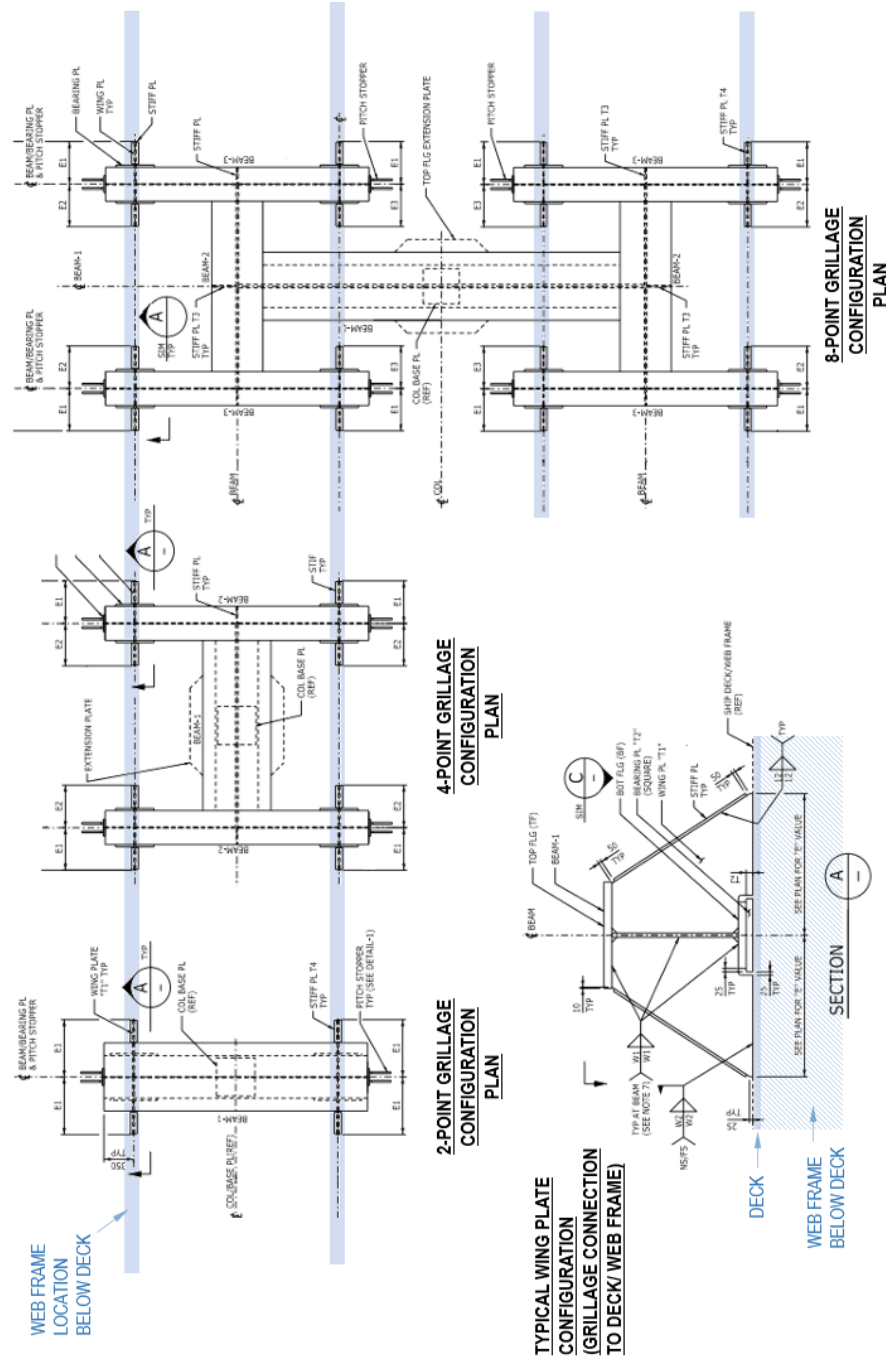


Figure 3 – Typical Grillage Configurations



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In cases where the module is lightweight, a skid type grillage and sea-fastening may be considered to minimize the time spent by the cargo-ship in the port. The concept is as shown below.



Figure 4 – Skid Grillage Configuration





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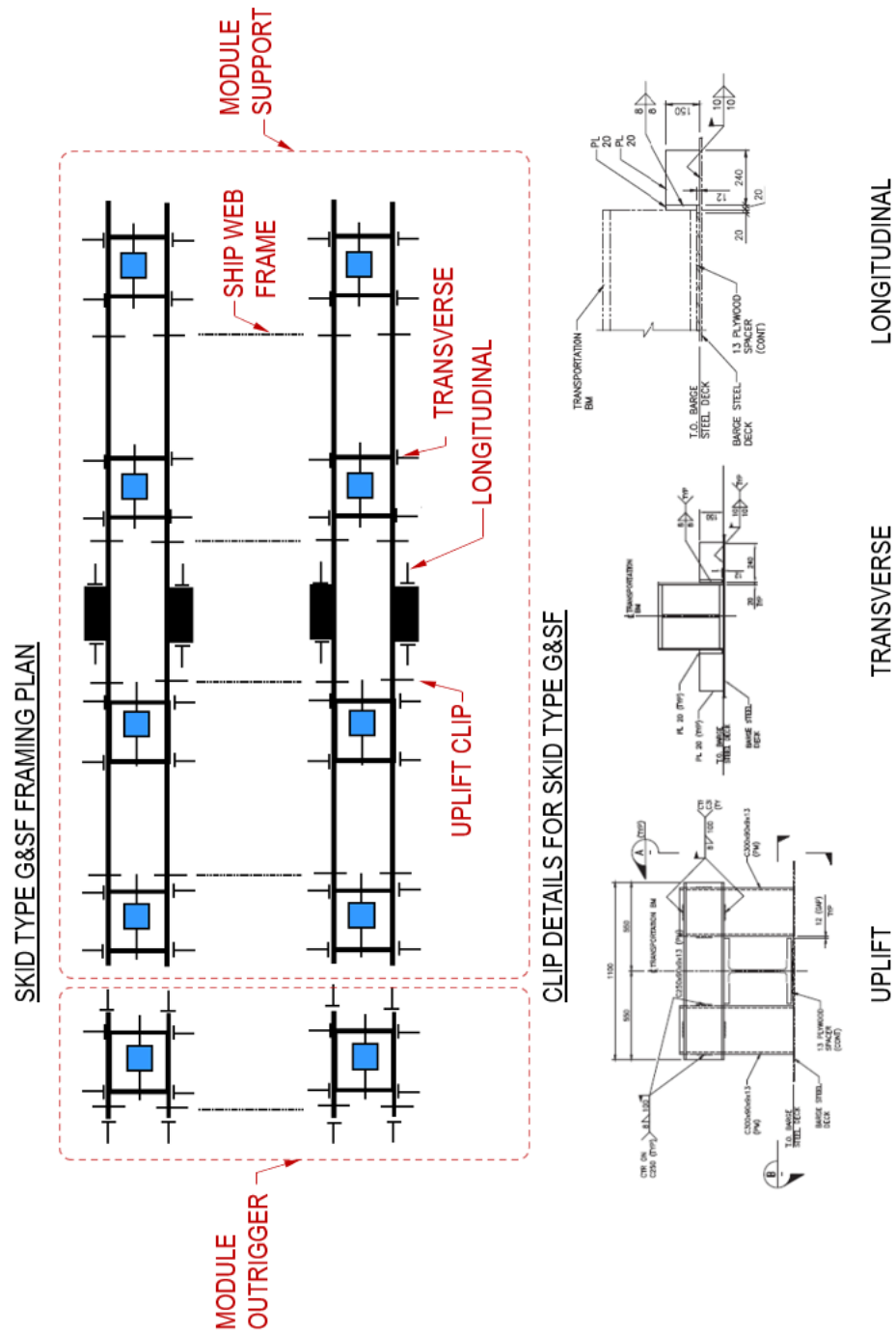


Figure 5 – Skid Grillage Framing Plan





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Sea fastening is installed after module set down and are typically the roll brace, pitch brace, shear stopper, and shear lug. A portion of the grillage beam wing plates may be installed after module set down where wing plates clash with the SPMT path. This condition should be avoided but, if such a clash can only be solved with the post set down installation of wing plate components, then a corresponding staged loading analysis should be completed.

Staged loading should also be considered for the initial deflection of the grillage frames when the module is set down and occurs prior to the installation of any sea fastening (such as roll and/or pitch braces). Software capable of completing the two-stage analysis (such as STAAD Pro Offshore) should be considered for completing this analysis.

Other types of sea-fastening are acceptable and should be configured in a manner to prevent a coupled action between the module and the ship frame. A generic assembly is shown in the figure below:

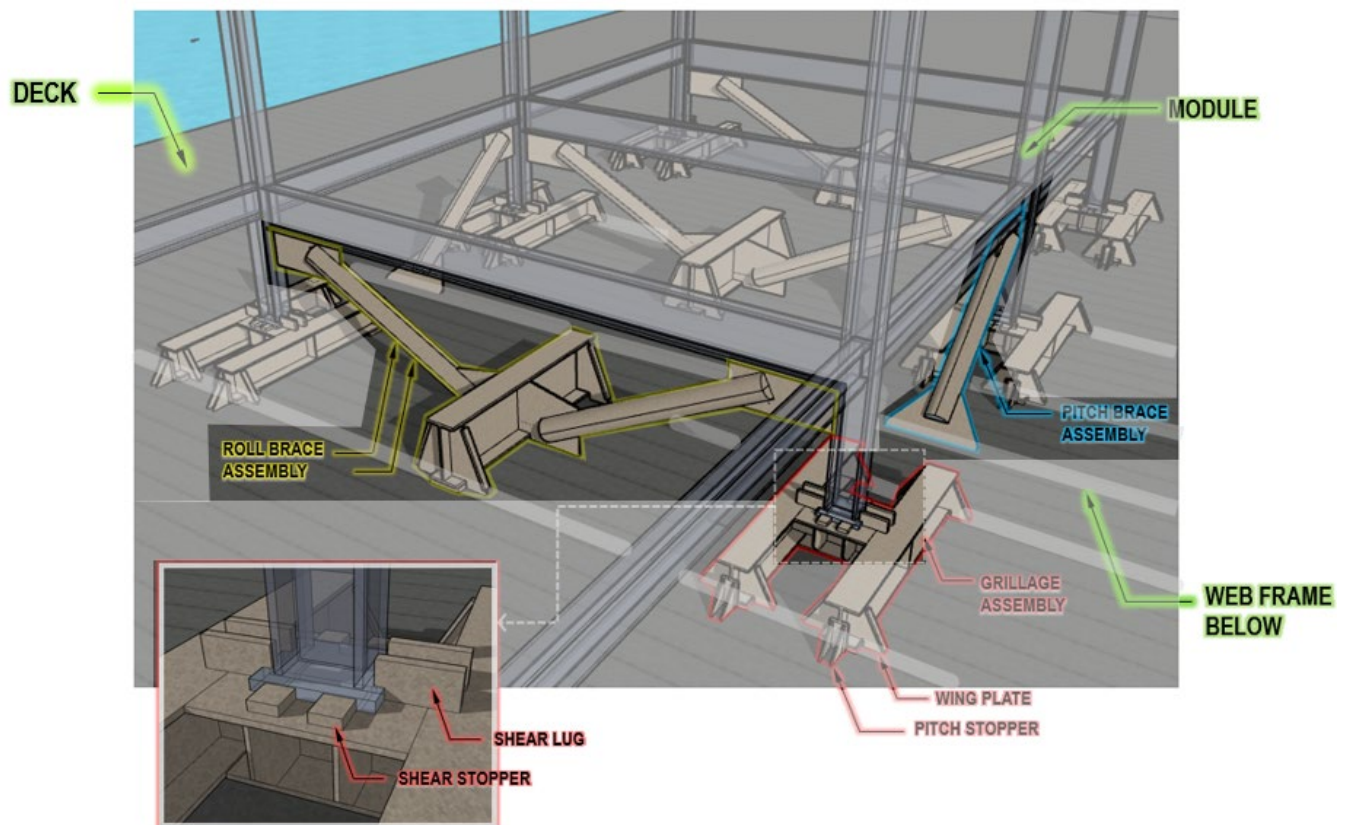


Figure 6 – Generic Grillage and Sea-Fastening Assembly



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Sea fastening has either a positive (welded) connection or are built just next to the cargo (module) such as stoppers and shear lugs. Hence, sea fastening components can be installed only after module set down and are primarily intended to hold the cargo in place during lateral movement and lateral loading like wind, and to resist any uplift forces induced by the marine transportation.

Pitch braces/ stoppers are to be installed in a single longitudinal location relative to the ship and preferably located in the middle of the module. Pitch braces located in 2 or more longitudinal locations relative to the ship would induce a coupled reaction between the ship and the module and require horizontal deformation due to sag and hog to be considered in the module design.

Generally, roll braces can be installed in 2 or more locations relative to the transverse dimension of the ship as ship deformation in the transverse plane is considered minimal. For the roll brace assembly shown below, the supporting beam may be evaluated using a two-stage analysis. The two-stage load distribution reduces the transfer of vertical load from the roll brace to the deck, since this vertical load is already being transferred from the columns to the grillage upon initial set down of the module. Potentially, the beam supporting the roll braces spanning between web frames along the deck may be eliminated since its primary function is to provide a load path for vertical load from the roll braces to the web frames.

Alternatively, the column spacing (and corresponding rolls braces) can be designed to align with the web frame spacing, providing a load path for any resultant vertical forces transferred from the roll braces into the deck without the additional beam needed to span between web frames. However, specific requirements should be coordinated with the ship owner and MWS to avoid coupled behavior.

Typical pitch, roll brace, shear stopper and shear lug construction as shown below:

PITCH BRACE AND COMPONENTS

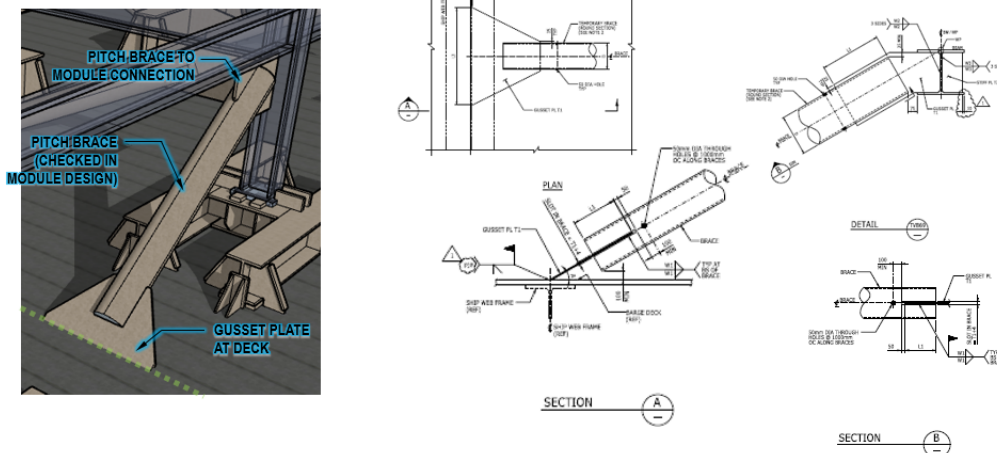


Figure 7 – Generic Pitch Brace Assembly



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#### ROLL BRACE ASSEMBLY COMPONENTS

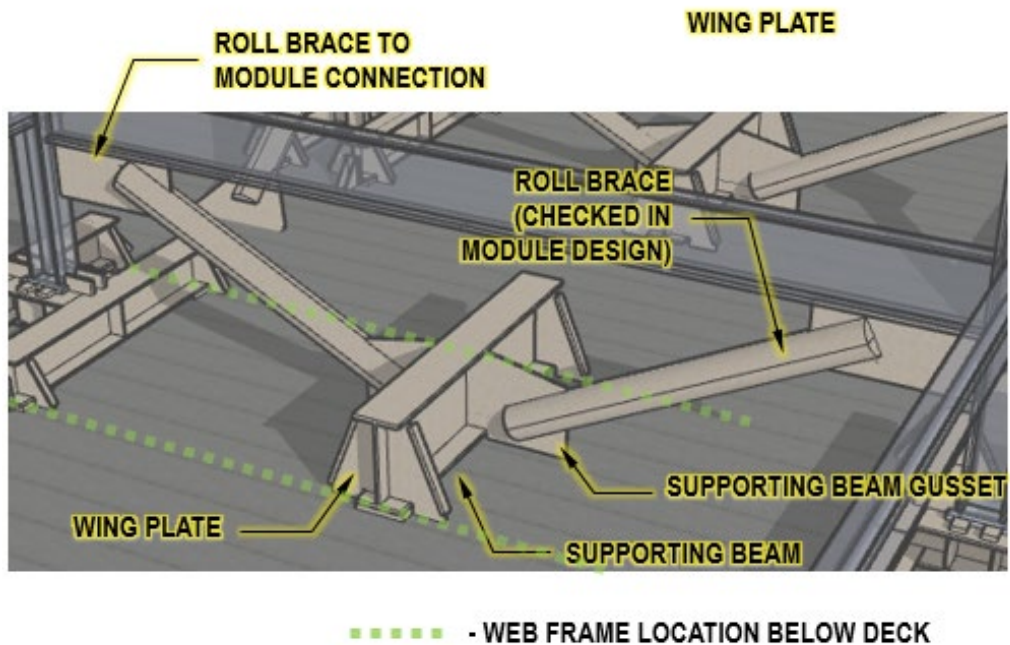


Figure 8 – Generic Internal Roll Brace Assembly



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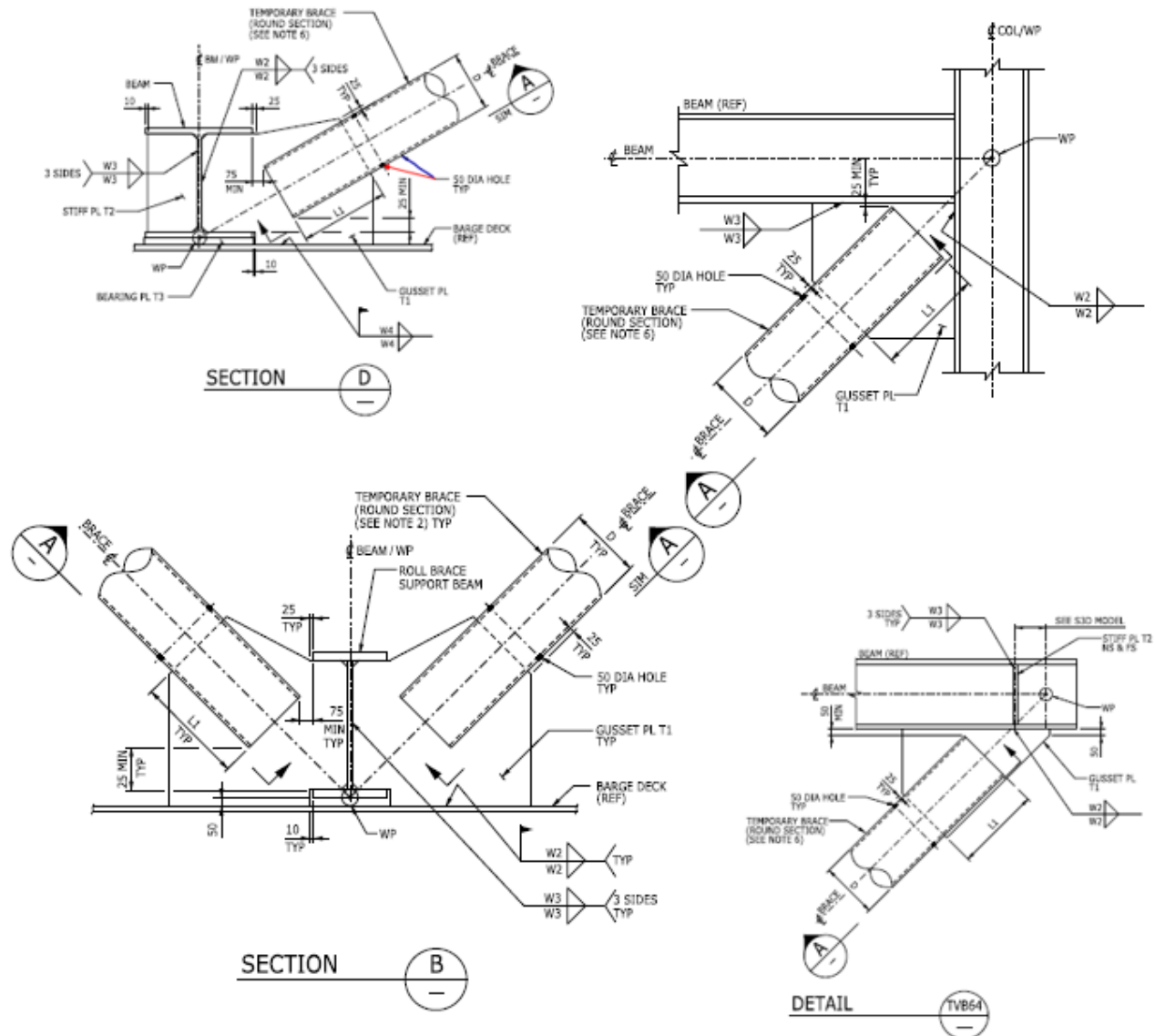


Figure 9 – Generic Roll Brace Details



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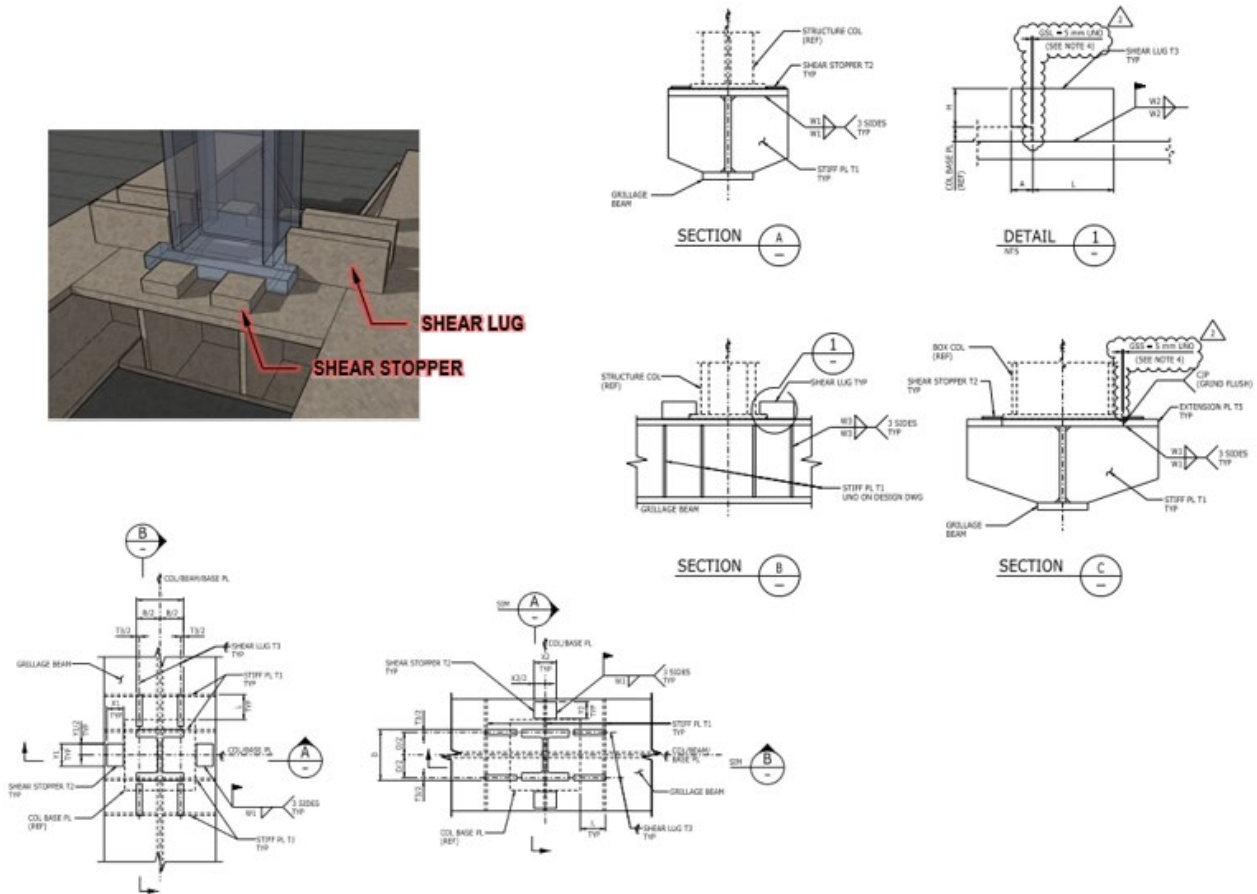


Figure 10 – Generic Shear Stopper and Shear Lug Assembly

#### 2.4 Input data

##### 2.4.1 Ship Frame Drawings (from ship owner)

It is essential to establish the location of hard points in the ship deck so that a capable load path can be conceptualized, analyzed, designed, and properly detailed for construction. The primary source of information for the location of ship web frames (hard points supporting the deck) is the ship frame drawings.

Note that ship frame drawings may vary from actual construction and a horizontal tolerance of  $\pm 50\text{mm}$  for web frame location is typically adopted and should be considered in the design. Ship Frame drawings are to be checked with an actual survey of web frames in the installation stage.





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#### 2.4.2 Ship Web Frame Capacities (from ship owner)

Hard points or support points have limiting capacities and are established and provided by the ship owner. The most common support points accessible from the deck are the web frames. Other support points include the bulk heads, ship shell and deck beams. Deck beams are of very low capacities and are usually not used. A sample framing plan with capacities are as shown below:

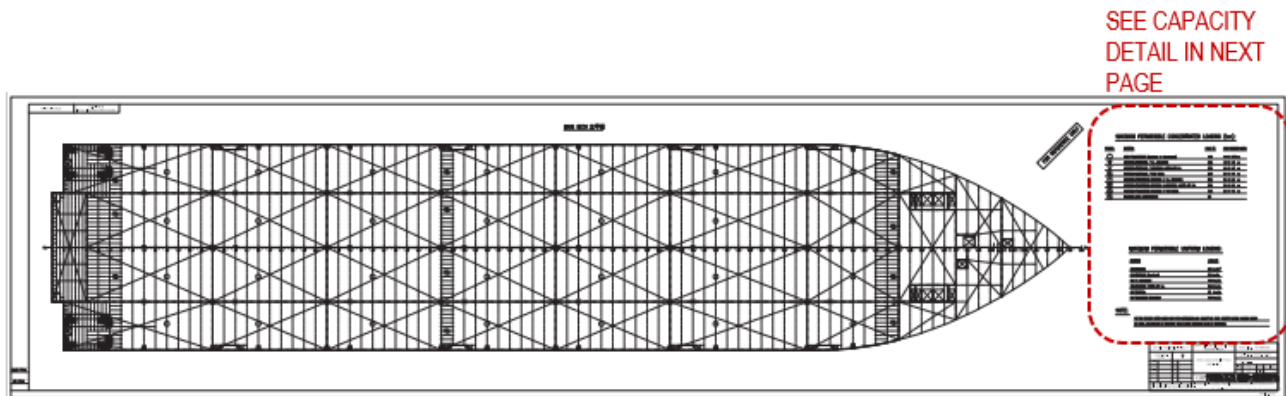


Figure 11 – Sample Ship Frame Framing Plan with Capacity Detail



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#### MAXIMUM PERMISSIBLE CONCENTRATED LOADING (ton):

| SYMBOL | POSITION                                               | LOAD (t) | LOAD BEARING AREA |
|--------|--------------------------------------------------------|----------|-------------------|
| ○      | DECK TRANSVERSES (Anywhere on transverses)             | 100      | 1.5 X 0.75 m      |
| ▽      | CROSSING WEBFRAME / C.L. BULKHEAD                      | 240      | 1.5 X 1.5 m       |
| ×      | CROSSING WEBFRAME / BULKHEAD 11375 OFF C.L.            | 270      | 2.0 X 1.5 m       |
| ⊠      | CROSSING WEBFRAME / SIDE SHELL                         | 255      | 2.0 X 1.5 m       |
| □      | CROSSING TRANSVERSE BULKHEAD / C.L. BULKHEAD           | 600      | 3.0 X 2.0 m       |
| ⊠      | CROSSING TRANSVERSE BULKHEAD / BULKHEAD 11375 OFF C.L. | 600      | 3.0 X 2.0 m       |
| ⊠      | CROSSING TRANSVERSE BULKHEAD / SIDE SHELL              | 405      | 2.0 X 1.5 m       |
| ⊠      | MIDSPAN DECK LONGITUDINALS                             | 25       |                   |

#### MAXIMUM PERMISSIBLE UNIFORM LOADING:

| POSITION                   | LOAD (t)              |
|----------------------------|-----------------------|
| ON MAINDECK                | 25 ton/m <sup>2</sup> |
| ON WEBFRAME (Anywhere)     | 60 ton/m              |
| ON C.L. BULKHEAD           | 100 ton/m             |
| ON BULKHEAD 11375 OFF C.L. | 100 ton/m             |
| ON TRANSOM                 | 25 ton/m              |
| ON TRANSVERSE BULKHEAD     | 150 ton/m             |

#### NOTE:

- 1) THE FIGURES LISTED HERE ONLY FOR REFERENCE AND CANNOT BE USED DIRECTLY WHEN LOADING CARGO.
- 2) DETAIL CALCULATION IS REQUIRED WHILE CARGO SECURING PLAN IS PREPARED.

Figure 12 – Sample Ship Frame Capacity Detail





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#### 2.4.3 Module Configuration (from module engineer)

The Cargo or Module design is not part of this guideline. It is assumed to be designed by the module engineer and should comply with all applicable project standards and specifications. The module is the load supported by the grillage and sea fastening and should have consistent geometry within project tolerances.

The basis for module geometry is:

1. Project Model – Electronic data on module configuration and geometry that are reliable and no longer subject to change. Typically, a steel model that was already issued for fabrication can be used as reference module geometry to design the grillage and sea fastening components.
2. Module Engineering Sketch/ Model Computer Model – When the project model is not yet final, the preliminary module geometry can be established from information supplied by the module engineer comprising primarily of the module engineering sketch or the model computer model. The engineering sketch is the preferred preliminary data between the two.

#### 2.4.4 Module Load from Computer Analysis (Envelope and Batch)

Grillage, Roll and Pitch brace bottom supports are modelled either as pin supports or as sliding supports (with vertical restraints) depending on the load conditions as outlined in Section 4 of this document. The reactions generated from the computer model analysis comprise the loads considered for grillage and sea fastening design. The support reactions from the complete analysis of harbor conditions and sailing conditions are consolidated and summarized in a table. The envelope reactions are used for simplicity; however, they can be broken down into batch reactions if needed for design optimization.

The module analysis software and design methodology (as well as any software limitation workarounds) used to obtain the G&SF design loads should be endorsed by the Marine Warranty Surveyor before the start of detailed engineering of phase to avoid engineering rework.

Specific software capabilities should include the following:

- Multi-Stage Loading Analysis and Design
- Auto-generation of marine acceleration loads
- Auto generation of COG shift

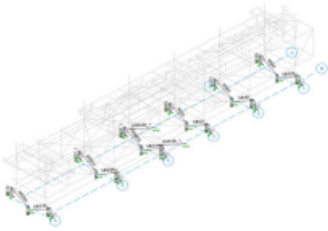
A sample grillage design force table is shown below:



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Module ID: XXXXXXXXXX  
Sketch:



| Grid     | 1    | 2    | 3    | 4    | 5    | 6    | 7 | 8 | 9 | 10 | 11 | 12 | Pitch Br |
|----------|------|------|------|------|------|------|---|---|---|----|----|----|----------|
| Roll Br  | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| B        | 670  | 718  | 767  | 707  | 859  | 470  | - | - | - | -  | -  | -  | 161      |
| Internal | 43   | 72   | 78   | 48   | 100  | 25   | - | - | - | -  | -  | -  | -        |
| A        | 682  | 789  | 816  | 715  | 866  | 438  | - | - | - | -  | -  | -  | 163      |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| Roll Br  | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| B        | -331 | -307 | -318 | -353 | -342 | -206 | - | - | - | -  | -  | -  | -153     |
| Internal | -29  | -25  | -28  | -32  | -50  | -30  | - | - | - | -  | -  | -  | -        |
| A        | -333 | -301 | -309 | -355 | -341 | -207 | - | - | - | -  | -  | -  | -156     |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| Roll Br  | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| B        | 183  | 175  | 178  | 188  | 214  | 153  | - | - | - | -  | -  | -  | -        |
| Internal | 204  | 220  | 192  | 180  | 206  | 183  | - | - | - | -  | -  | -  | -        |
| A        | 185  | 173  | 185  | 190  | 214  | 154  | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| Roll Br  | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| B        | 0    | 0    | 0    | 0    | 0    | 0    | - | - | - | -  | -  | -  | 322      |
| Internal | 0    | 0    | 0    | 0    | 0    | 0    | - | - | - | -  | -  | -  | 325      |
| A        | 0    | 0    | 0    | 0    | 0    | 0    | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| Roll Br  | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| B        | 214  | 270  | 326  | 287  | 390  | 210  | - | - | - | -  | -  | -  | 27       |
| Internal | 8    | -    | 15   | 25   | -    | 18   | - | - | - | -  | -  | -  | -        |
| A        | 208  | 296  | 325  | 253  | 386  | 164  | - | - | - | -  | -  | -  | 26       |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| Roll Br  | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| B        | 0    | 0    | 0    | 0    | 0    | 0    | - | - | - | -  | -  | -  | -16      |
| Internal | -4   | 0    | -13  | -13  | 0    | -15  | - | - | - | -  | -  | -  | -        |
| A        | 0    | 0    | 0    | 0    | 0    | 0    | - | - | - | -  | -  | -  | -18      |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| Roll Br  | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| B        | 41   | 54   | 63   | 51   | 77   | 37   | - | - | - | -  | -  | -  | -        |
| Internal | 11   | 0    | 26   | 37   | 0    | 29   | - | - | - | -  | -  | -  | -        |
| A        | 42   | 59   | 65   | 47   | 77   | 31   | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| Roll Br  | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |
| B        | 41   | 54   | 63   | 51   | 77   | 37   | - | - | - | -  | -  | -  | 53       |
| Internal | 42   | 59   | 65   | 47   | 77   | 31   | - | - | - | -  | -  | -  | 52       |
| A        | 42   | 59   | 65   | 47   | 77   | 31   | - | - | - | -  | -  | -  | -        |
| 0        | -    | -    | -    | -    | -    | -    | - | - | - | -  | -  | -  | -        |

Figure 13 – Sample Reactions table for grillage load consideration

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[illegible]

#### 2.4.5 Stowage Plan (From Logistics Group)

Civil Structural Architectural



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A sample portion of the stowage plan is as shown below:

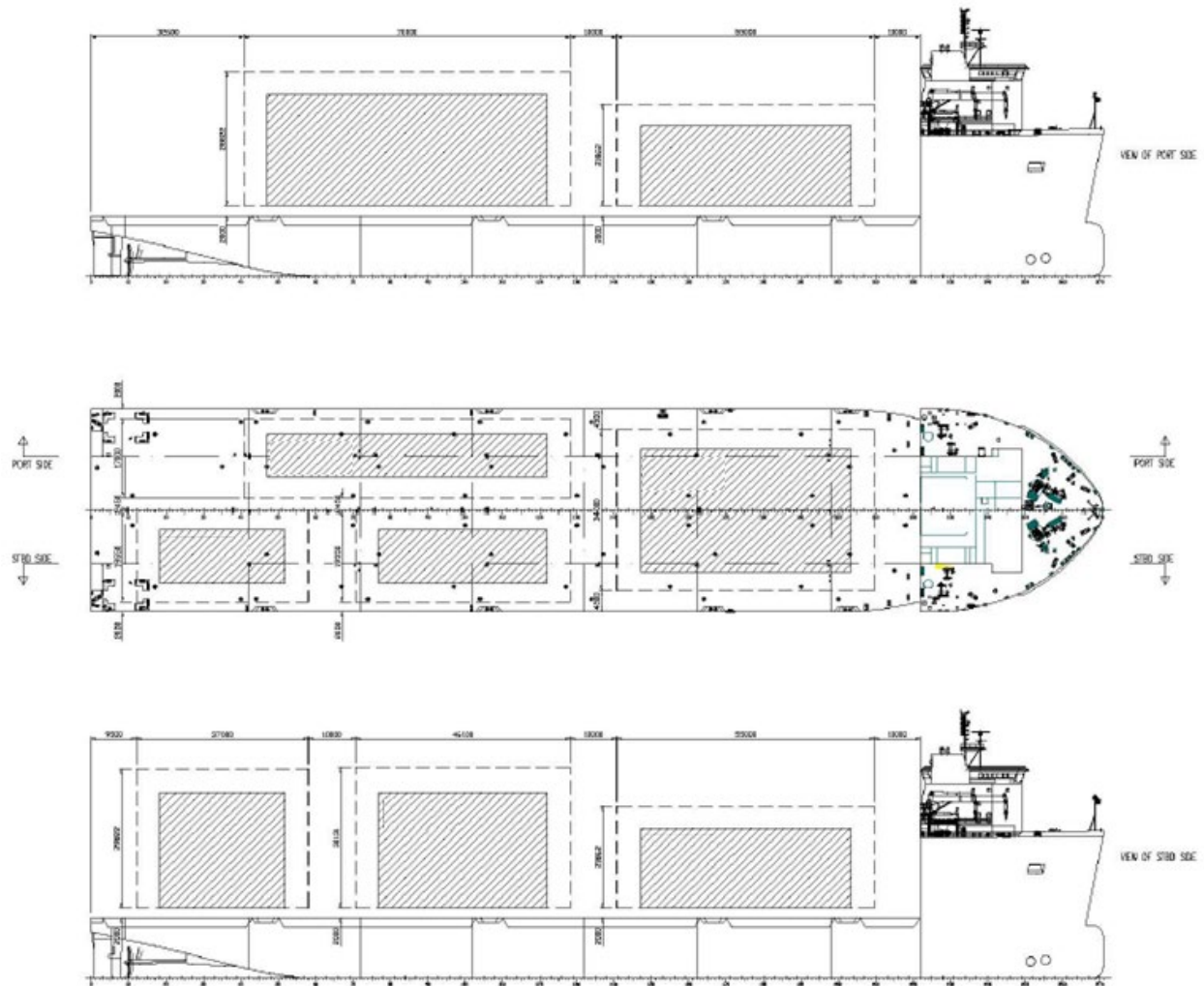


Figure 15 – Sample Stowage Plan for reference in the grillage design.

#### 2.4.6 SPMT Transport Arrangement Drawings (From Heavy Haul Contractor).

Self-propelled modular transporter (SPMT) configuration is designed by the Heavy Haul Contractor to carry the module in a specific manner and position and is documented in the SPMT Transport Arrangement drawing. The geometry of the proposed grillage is to



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be checked for any clash with respect to travel path of the SPMTs. A clearance of 300mm is typically maintained between the extent of the grillage and path of the SPMT. Any encroachment with this clearance can be managed in several ways:

1. Consider installation of some wing plates after set down.
2. Consider changing the location of the grillage.
3. Consider changing the dimensions of the grillage.
4. Consider changing SPMT locations in collaboration with the Heavy Haul Contractor.

#### 2.4.7 Typical Grillage Vertical Load Capacity (Initial evaluation).

Depending on the availability of ship web frame capacities which are the typical limiting factor for the grillage design, an initial limiting load capacity per grillage type (2-point, 4-point, 6-point and 8-point) can be developed considering vertical loads only from the computer model analysis reactions table. This will be affected to a certain degree by the induced lateral loads, however if a suitable margin of safety is maintained between the allowable vertical load for a grillage frame and the web frame capacity, then it is reasonably likely that a suitable grillage configuration was chosen. Care must be done to maintain an economic design and avoid using an excessive margin of safety.

#### 2.5 Preliminary Grillage Framing Plan

The preliminary grillage framing plan is one of the more important steps in developing the grillage and sea fastening design. It is essential that judgement and creativity be exercised, as this will affect the load distribution and eventually the economy and feasibility of the design. The following items are to be checked:

1. Preliminary limiting capacity of proposed grillage
2. Clash with stowage plan boundaries
3. Clash with SPMT clearance requirements
4. Clash of SPMT with ship
5. Constructability concerns (such as available clearance for welding, weight of pieces for handling, stroke height of the SPMT, etc.)
6. Availability of supporting web frames for pitch braces including the acceptability of pitch brace angle with respect to the deck and module frame.

#### 2.6 Analysis and design of components

Note that it should be decided at the beginning of the design process whether the grillage and sea fastening will be reused, as this has a significant impact on grillage design (i.e., fatigue considerations) and ship / module sail and return schedule.

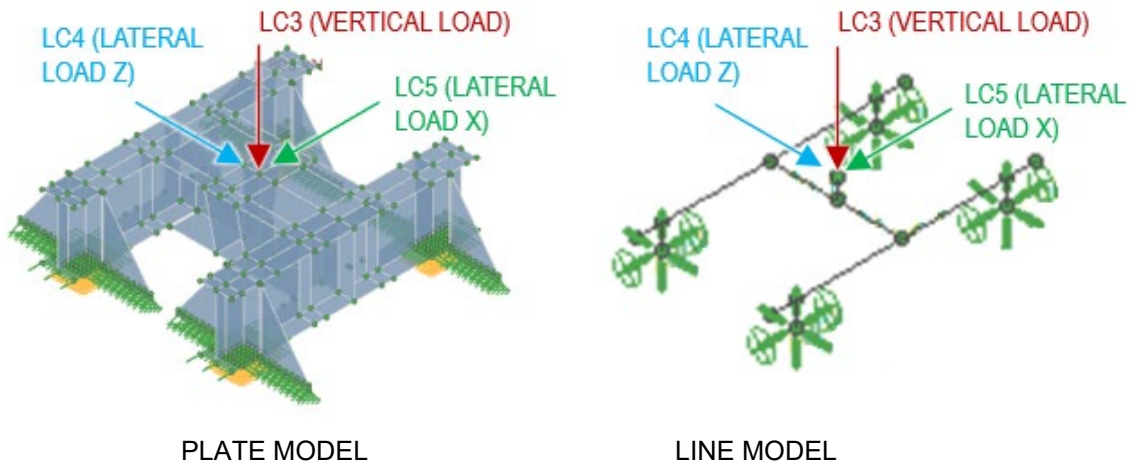
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**STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES**

**10 – Grillage and Sea Fastening Design**

**2.6.1 FEM using plate elements to characterize behavior and simplify analysis.**

A plate FEM model of the grillage would be an accurate characterization of elastic behavior. However, it is laborious if done for every grillage to be analyzed and designed. Hence a sample 4-point grillage was modelled and analyzed using plate elements and compared to the results of a FEM model using line elements. The RISA 3D comparative modelling and summary of results are as follows.



**Figure 16 – Beam Strong Axis Bending Stresses from Vertical Loads**





## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design

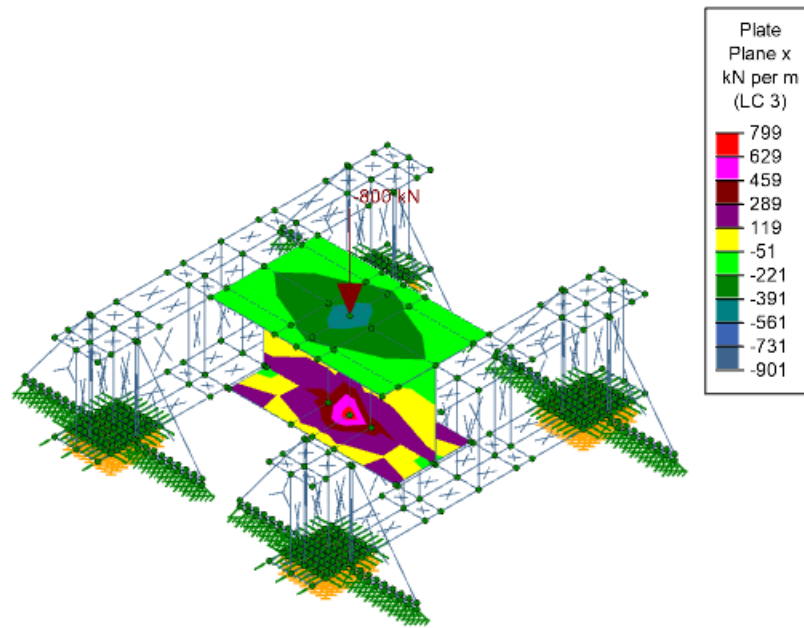
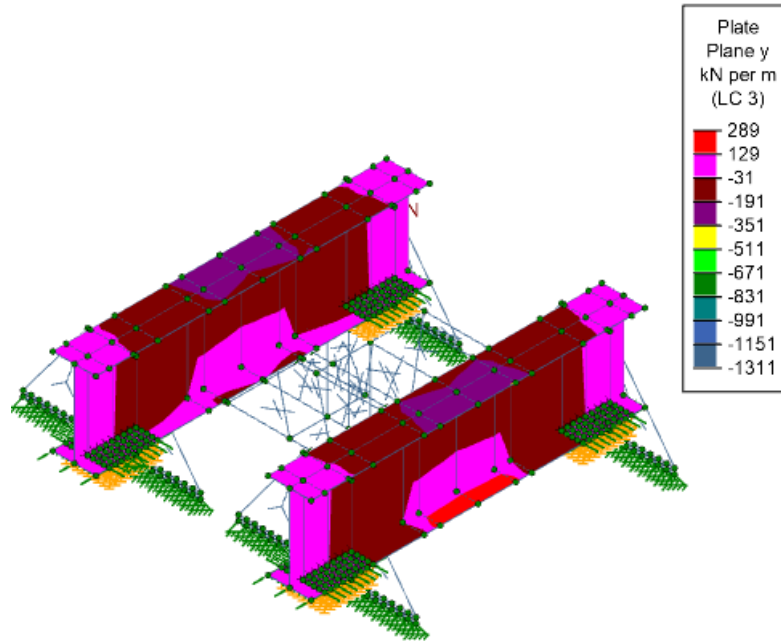


Figure 17 – Primary Beam Bending Stresses

## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design



**Figure 18 – Secondary Beam Bending Stresses**

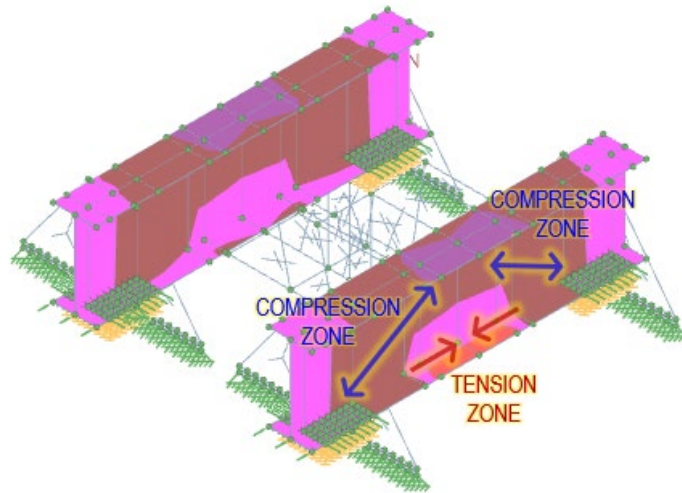
#### Observations:

- The beam total flange forces in the FEM plate element model are smaller than, but similar to, the total flange force in the line element model using the location of highest moment and with flange force from the line element model given as follows:
- $$Flange\ Force = Moment \times \frac{depth-flange\ thickness}{2 \times I_x} * b * t_f$$
- Thus, for flange stress evaluation, the line element model is conservative but reasonably close to the plate element model.
- The primary beam in the plate element model follows a stress distribution that is similar to the bending behavior wherein the stress varies linearly throughout the depth of the section. This is consistent with the assumed line element stress distribution wherein the basic assumption is that plane section remains plane in bending.
- For secondary beam, the stress distribution in the plate element model follows a compression and tension zones like a tie-strut model and is as illustrated below:

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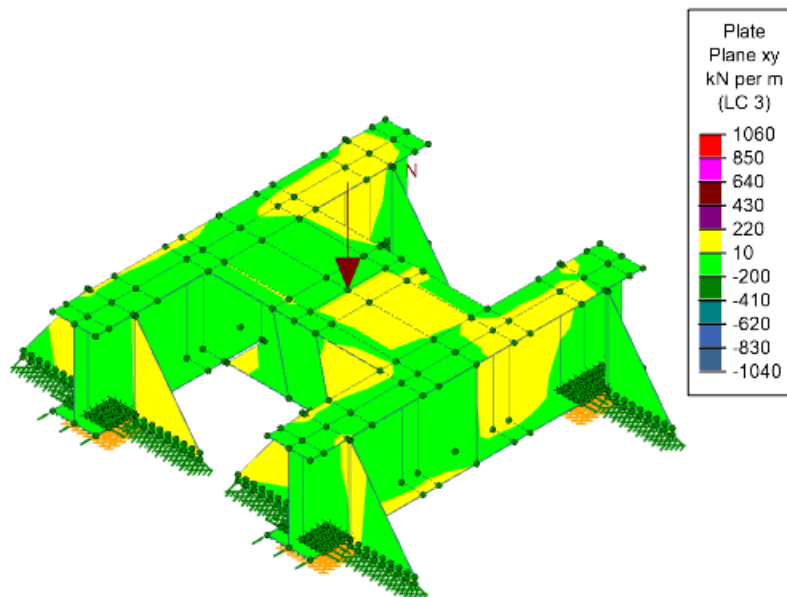
**STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES**

**10 – Grillage and Sea Fastening Design**



**Figure 19 – Secondary Beam Stress Distribution**

The use of shear evaluation for compression strength in the web is a conservative approach. The compression capacity of the web is much higher than its shear capacity.



**Figure 20 – Beam Strong Axis Shear Stresses from Vertical Loads**



## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design

#### Observations:

The shear stress distribution process in the web is similar to the shear distribution pattern in line element analysis, however the value derived is smaller in the plate element analysis and is indicative of plate mechanisms like the strut-tie effect mentioned above. It is thus conservative to analyze the beams as line elements.

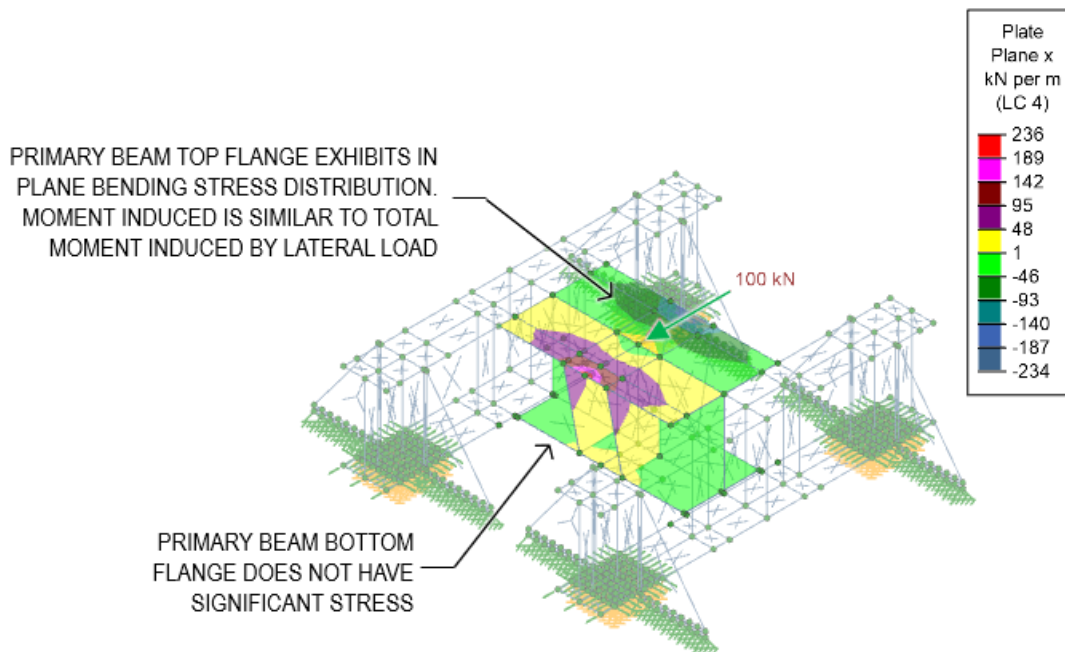
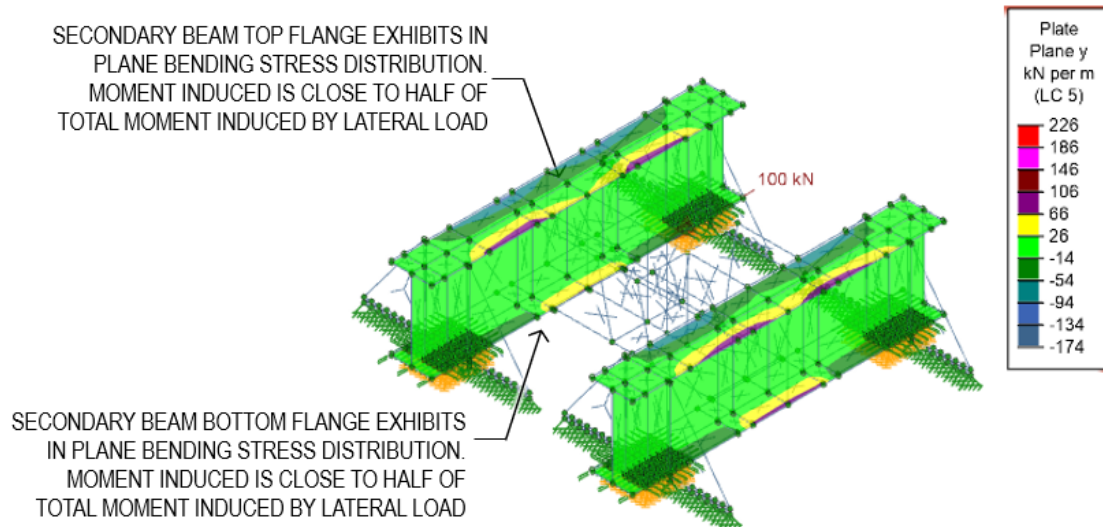


Figure 21 – Beam Weak Axis Primary Bending Stresses from Lateral Loads



## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design



**Figure 22 – Beam Weak Axis Primary Bending Stresses from Lateral Loads**

**Observations:**

For Primary beams, the bending stress distribution is like the bending stress distribution pattern in line element analysis, however only the top flange is engaged.

For Secondary beams, the bending stress distribution is like the bending stress distribution pattern in line element analysis where both the top flange and bottom flange contribute almost equally to the resistance.



## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design

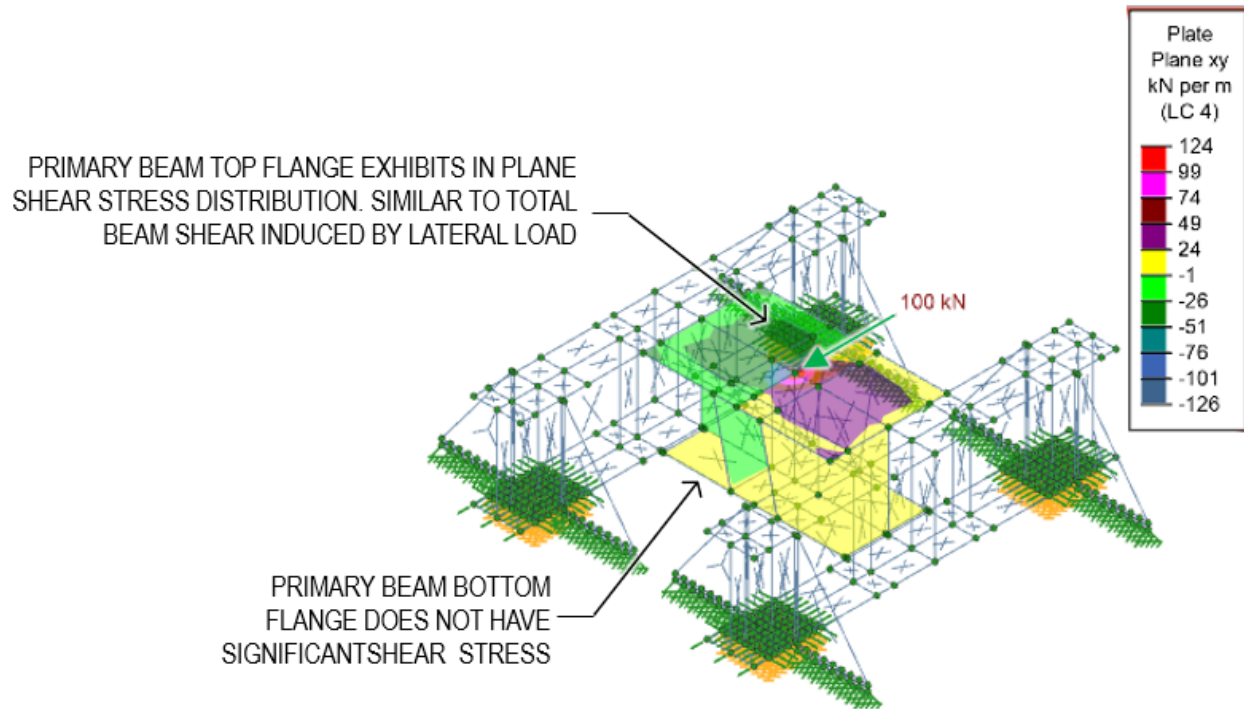
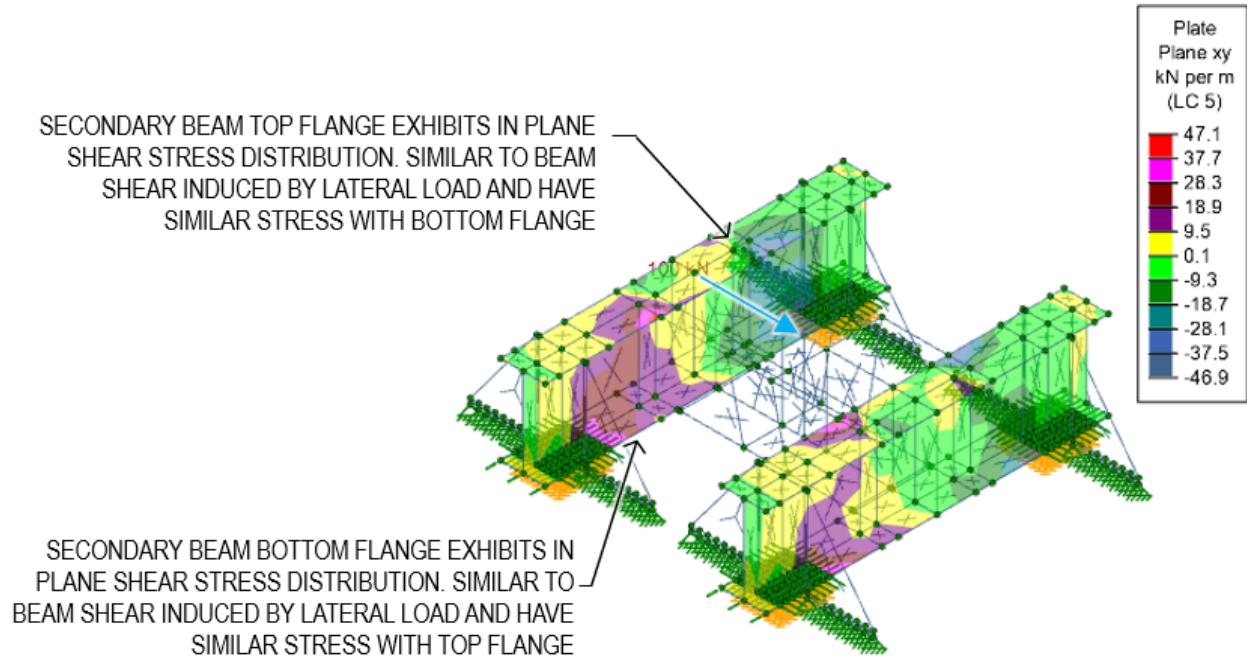


Figure 23 – Primary Beam Weak Axis Shear Stresses from Lateral Loads

## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design



**Figure 24 – Secondary Beam Weak Axis Shear Stresses from Lateral Loads**

#### Observations:

The primary beam has a large top flange width relative to its span. It exhibits diaphragm action and attracts almost all the lateral shear load applied and spares the bottom flange from any stress.

The secondary beam receives the load not at the top, but from the primary beam web where it is distributed almost evenly between the top and bottom flanges.

#### Overall Summary:

For Vertical loads, the results of the line element modelling are reasonably accurate in comparison to FEM with plate elements. The resulting stresses evaluated in the code check are higher and thus the UC results are conservative.

For Lateral loads, the following adjustments need to be done for the primary beam only considering that the top flange acts as a diaphragm and absorbs almost all the loads.

- Weak axis bending UC to be multiplied by the ratio of section weak axis plastic modulus divided by top flange plastic section modulus.

$$UC_{wk\_axis\_adjusted} = UC_{wk\_axis\_software} \times \frac{Z_{x\_all}}{Z_{x\_topflange}}$$



## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design

Combined UCs should be recomputed on manual templates to consider the adjusted weak axis bending UC.

- Weak axis shear UC to be multiplied by the ratio of total flange area divided by top flange area.

$$UC_{wk\_axis\_adjusted} = UC_{wk\_axis\_software} \times \frac{A_{v\_all\_flange}}{A_{v\_topflange}}$$

Combined UCs should be recomputed on manual templates to consider the adjusted weak axis bending UC.

- Axial compression or tension UC to be multiplied by the ratio of total area divided by top flange area.

$$UC_{adjusted} = UC_{software} \times \frac{A_{g\_all\_flange}}{A_{topflange}}$$

Combined UCs should be recomputed on manual templates to consider the adjusted weak axis bending UC.

#### 2.6.2 Mathematical model of grillage for element evaluation and design

The grillage frame elements are to be analyzed against the loads from the reaction table. Grillage components are to be modelled as line elements in a computer model according to the following configuration:

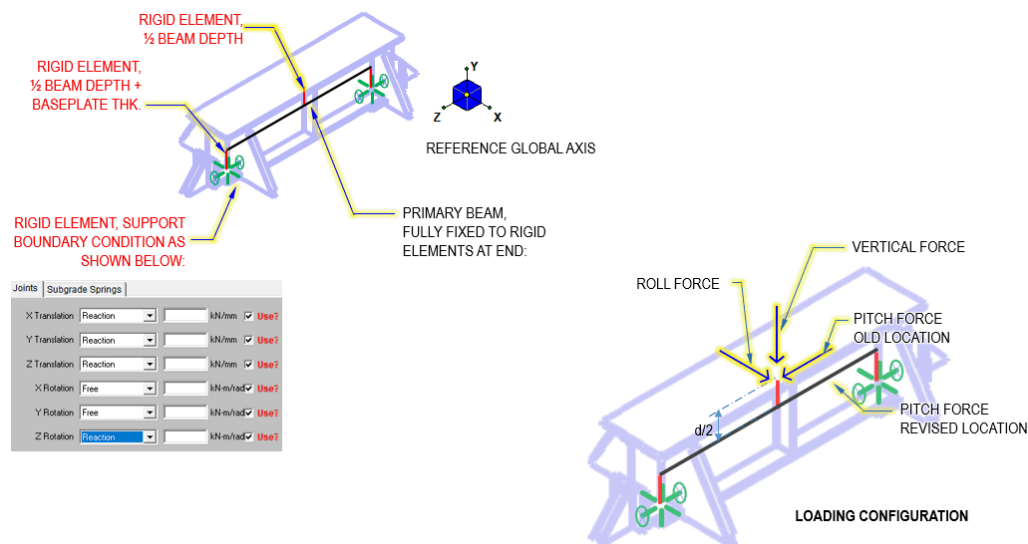


Figure 25 – 2-Point Grillage Modelling Guide



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### 10 – Grillage and Sea Fastening Design

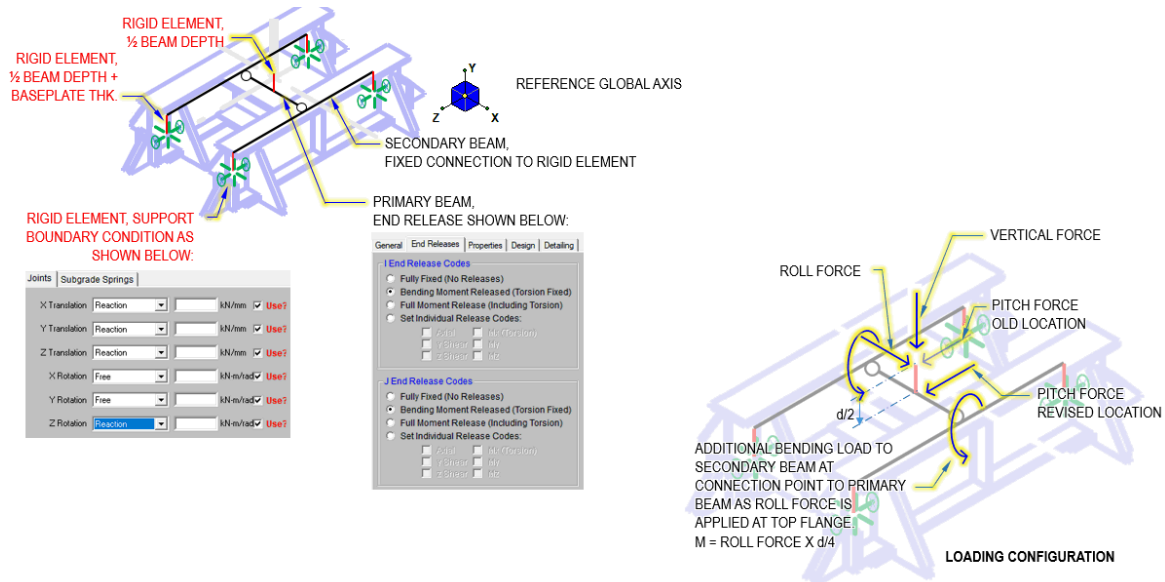


Figure 26 – 4-Point Grillage Modelling Guide

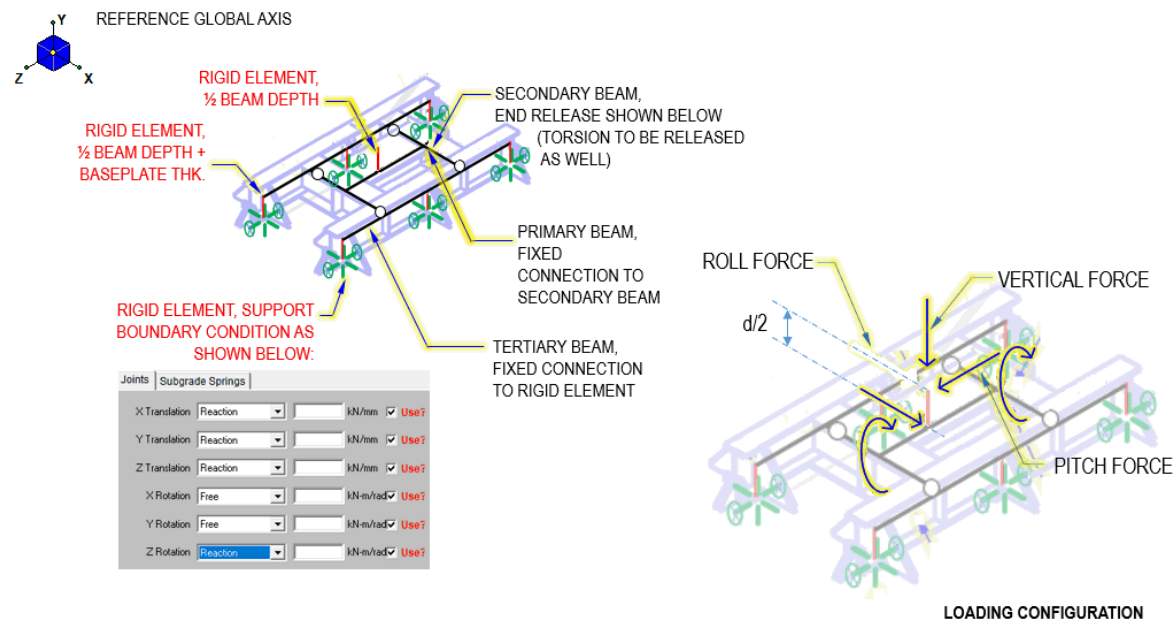


Figure 27 – 6-Point Grillage Modelling Guide



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### 10 – Grillage and Sea Fastening Design

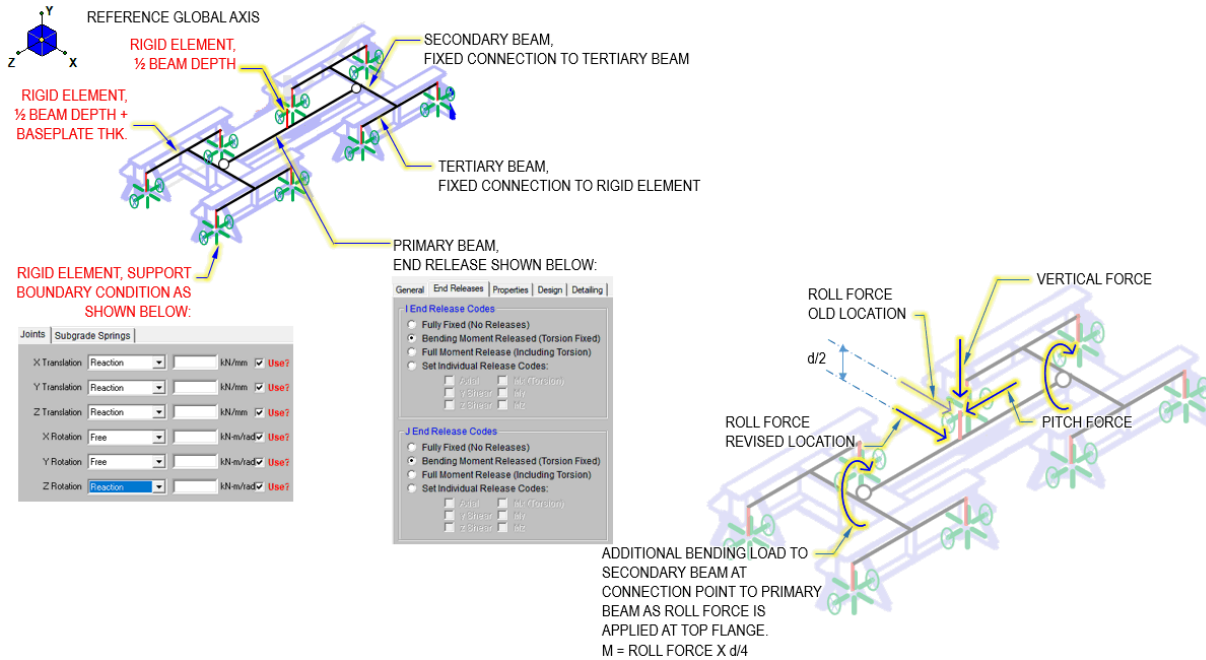


Figure 28 – 8-Point Grillage Modelling Guide

Note that the UC evaluation is to be adjusted as discussed in Section 2.6.1. For connection design, all lateral loads in the primary beam are to be taken by the primary beam top flange connections to the secondary beams.

#### 2.6.3 Reaction loads for wing plate / web frame loads analysis

The computer model for the grillage will be analyzed and will produce reaction loads representing the load that will be transferred to the ship web frame via the wing plate and baseplate assembly including the pitch stoppers at ends. These reactions are to be analyzed manually and checked if the web frame and deck capacities are satisfied. A sample analysis table is shown below.

## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design

References: (1) AISC Manual 14th Ed.  
 (2) AISC Design Guide 29

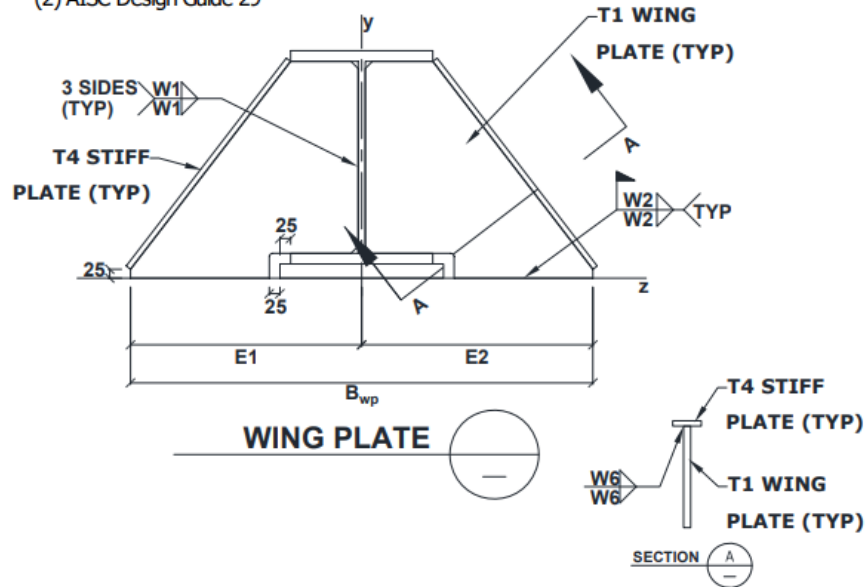


Figure 29 – Sample Standard Wing Plate Construction Detail

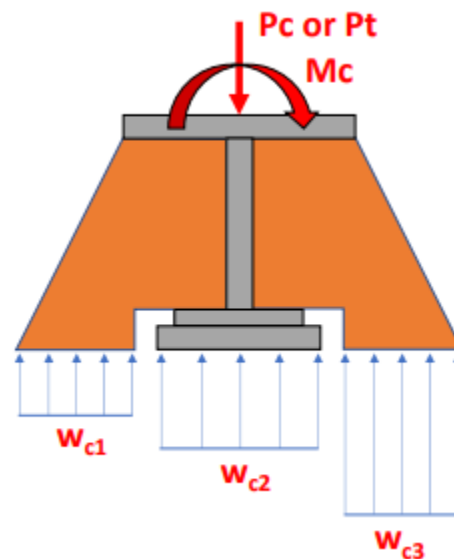


Figure 30 – Wing plate assumed distribution of stress at base  
 (Loads are from computer model reaction results)



## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design

Compression Case:

| Column | Beam Size           | Pc (kN) | M (kN-m) | W <sub>c1</sub> (t/m) | W <sub>c2</sub> (t/m) | W <sub>c3</sub> (t/m) | Remarks       |
|--------|---------------------|---------|----------|-----------------------|-----------------------|-----------------------|---------------|
| A1     | H800X1000X700_60X50 | 511.212 | 120.969  | 6                     | 32                    | 57                    | < 112 t/m, OK |
| B1     | H800X1000X700_60X50 | 503.212 | 119.769  | 6                     | 31                    | 56                    | < 112 t/m, OK |
| A2     | H800X1000X700_60X50 | 817.185 | 156.215  | 17                    | 50                    | 84                    | < 112 t/m, OK |
| B2     | H800X1000X700_60X50 | 746.191 | 157.803  | 13                    | 46                    | 79                    | < 112 t/m, OK |
| A3     | H800X1000X700_60X50 | 577.089 | 113.984  | 12                    | 36                    | 60                    | < 112 t/m, OK |
| B3     | H800X1000X700_60X50 | 544.423 | 109.885  | 10                    | 34                    | 57                    | < 112 t/m, OK |
| A4     | H800X1000X700_60X50 | 748.059 | 170.656  | 10                    | 46                    | 82                    | < 112 t/m, OK |
| B4     | H800X1000X700_60X50 | 740.060 | 168.927  | 10                    | 46                    | 81                    | < 112 t/m, OK |
| A5     | H800X1000X700_60X50 | 633.877 | 138.529  | 10                    | 39                    | 68                    | < 112 t/m, OK |
| B5     | H800X1000X700_60X50 | 629.211 | 138.500  | 10                    | 39                    | 68                    | < 112 t/m, OK |
| A6     | H800X1000X700_60X50 | 393.215 | 117.576  | -1                    | 24                    | 49                    | < 112 t/m, OK |
| B6     | H800X1000X700_60X50 | 419.880 | 116.915  | 1                     | 26                    | 51                    | < 112 t/m, OK |

Figure 31 – Sample Wing Plate Loads per grillage type

#### 2.6.4 Welded connection design

Connection design to follow AISC 360-10 and AISC connection design guide requirements. Direct connection to the deck including the welding of the wing plate, pitch and roll brace gusset, and grillage pitch stoppers shall be designed without the 1/3 overstrength factor in accordance with DNVGL-ST-N001 2016.

Note that the UC evaluation is to be adjusted as discussed in section 4.7.1. For connection design, all lateral loads in the primary beam are to be taken by the primary beam top flange connections to the secondary beams.

#### 2.6.5 Output data

The following documents are to be produced by the grillage and sea fastening engineer and are to be submitted to the required parties:

##### 1. Engineering Sketch

Sufficient information should be provided so that the grillage and sea fastening design is built as intended. As a minimum the following information should be provided:

- Material specification including welds, bolts if any, and other complementary components.
- Reference locations relative to ship including connection to ship frame location and connecting ship frame identification name, web-frame number, and ship gridline number.
- Module baseplate plan dimensions and elevations including module gridline IDs.
- Design details like weld dimensions, gusset plate dimensions, etc.
- Special instructions as necessary to include the following:



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## STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

### 10 – Grillage and Sea Fastening Design

- Construction sequence to avoid module set-down clash (i.e., SPMT path).
- Confirmation of Web frame locations.
- Confirmation of ship appurtenances than can clash with proposed grillage and sea fastening.

#### 2. Calculation document

Calculation document should be completed and submitted MWS in accordance to agreed-upon format.